RAY: Resilience Assistant for You

An Accountable GeoAI System for Place-Based Disaster Intelligence

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Abstract

Most GeoAI disaster systems demonstrate autonomy: a language model that runs a spatial pipeline. RAY (Resilience Assistant for You) demonstrates *accountability*: a risk-analyst agent that operates inside the *Steward Harness*, a technical layer that enforces outcome, process, and institutional validity during operation rather than promising them in prose. Every artifact is hashed into an append-only audit log; a declarative claim policy scopes what the agent may assert by role, evidence tier, spatial resolution, and geographic authority; a separate distribution policy scopes what a build may publish, verified by a CI gate that fails the deploy on violation; every factual sentence must cite an artifact identifier or be refused; and a *verifiability* axis (retained > re-derivable > cited-only) states what a reader can do to check a claim. We exercise the harness on a nationwide hourly hazard watch and three deep cases across two hazard types — the 2025 Eaton Fire, Hurricane Milton (2024), and Hurricane Ian (2022) — delivered through an installable, keyless web app with resident and planner modes. We then ask what it takes for a vision–language model's output to count as evidence. Six open VLMs, served locally on one 24 GB GPU, graded 1,225 street-view images and image pairs per model across the three cases under verbatim published prompts; every prediction is a hashed record, thresholds fail closed, and the resulting grids are flagged *model-derived*: publishable as an evaluation, unusable as a claim. Single-image grading comes within two points of closed commercial models; pre/post change grading does not, and each model fails with a different fixed class bias. The test suite doubles as a verifiable evaluation environment (363 Python and 67 app tests in CI), and the audit record preserves three incidents where the harness caught the system violating its own rules — evidence, we argue, that the mechanism does work that documentation alone does not.

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CCS Concepts

• **Information systems → Geographic information systems**; • **Computing methodologies → Artificial intelligence**.

Keywords

GeoAI agents, accountability, disaster resilience, social vulnerability, provenance, policy enforcement, vision–language models, H3

1 Introduction

Large-language-model agents can now operate geographic information systems: they select tools, write spatial queries, and produce maps and narratives with little human intervention [4, 7]. In disaster response this capability meets an unforgiving audience. An emergency manager acting on a tile-level damage estimate, or a resident reading a sentence about their own street, cannot audit a transformer. The prevailing answer is disclaimers — operational platforms ship forecasts with the caveat that they "complement, do not replace" official warnings — which locates accountability in the reader rather than in the system.

RAY locates it in the system. Following the framing of Yang and Zou [6], we treat an agent's claim as valid only when three conditions hold at once: the *outcome* passes executable spatial checks, the *process* that produced it is preserved and inspectable, and the claim is *institutionally* authorized — the right actor asserting the right thing at the right resolution. The *Steward Harness* enforces all three in code, and adds a fourth question that arises the moment third-party data enters: what can a *reader* do to check this, and may the evidence be kept at all?

This paper describes the deployed system[1] and reports what its enforcement actually caught. Our contributions are: (1) a harness architecture with two declarative policy planes — claim and distribution — that fail closed, plus a per-claim *verifiability* axis with weakest-link semantics; (2) three harness-built deep cases across two hazard types, joining structure-level damage ground truth, debris volumes, CDC social vulnerability, and cross-view street imagery on H3 r9 grids, behind a nationwide hourly watch; (3) a treatment of model output as *governed evidence*: six open vision–language models evaluated on 1,225 labelled samples per model across the three cases, with every prediction recorded, thresholds

[1] Live site, app, and repository: https://rayford295.github.io/ray-resilience/

that fail closed, and results that the agent is forbidden to cite; (4) a test suite designed as a verifiable evaluation environment, and an append-only audit record that preserves three self-caught violations — including one where the deployed site violated the project's own written distribution constraint and the fix was a new policy plane, not a patch.

2 The Steward Harness

The harness sits between every pipeline stage, and between the language model and every sentence it emits.

Outcome validity. Executable spatial checks run inside each build: CRS assertions, raster×vector and tract-join integrity, sanity bounds, and *mandatory* per-tile uncertainty fields — a grid cell without a declared uncertainty structure fails the build.

Process validity. Every registered artifact receives a SHA-256 digest and an append-only audit row naming its agent, inputs, and UTC timestamp. Failures are recorded, never erased; the app renders lineage from any map layer back to timestamped source snapshots. The gateway's own audit is redacted by one rule — it holds no finer location than the claim plane lets any answer use: a point is stored as its H3 r9 cell, an area's corners rounded to ~110 m, and the question as sha256 plus length, so a caller can prove their question produced a row without the log retaining what a resident typed about their own home.

Institutional validity. A single YAML policy file carries two planes of ordered, first-match rules with default deny, validated at construction time — an unknown match key fails loudly at load rather than silently broadening authorization. The *claim plane* scopes what may be asserted, by role (resident vs. planner), evidence tier, spatial resolution, and geography: tile-level evidence never yields parcel-level claims. The *distribution plane* scopes what a build may *publish*, by artifact class, resolution cap, and audience; CI verifies the assembled public site against it and fails the deploy on violation. One `published_events` list feeds both planes, so the agent cannot read what the publisher withholds. Section 5 explains why the second plane and that shared list exist.

Citation by default. Every factual sentence the agent produces must cite an artifact identifier. The exemption set is closed (questions, imperative safety advice, statements about what the answer cannot say), so a sentence form nobody anticipated costs a refusal, not an uncited claim. Refusals name the policy rule that triggered them.

Verifiability. Evidence tiers describe depth; they do not say whether a reader can *check* a claim. Each artifact therefore carries `verifiability` ∈ {`retained`, `re-derivable`, `cited-only`} with weakest-link aggregation, and a `license` attribute gates redistribution by rule. The axis earns its keep where retention is forbidden: map-platform terms disallow storing map content, which is structurally incompatible with proving traceability by hashing inputs. The harness resolves this by attesting to the *request* plus a response digest — a lookup record that is publishable precisely because it holds no content — and a containment property test enforces that no response payload ever reaches the audit record. Grounded free-text generation is not reproducible even at temperature zero, so it falls through to default-deny: the citable unit is the citation, never the prose. A fourth value, `open-license-attribution`, is exercised in production for OSM-derived facility context (ODbL): redistributable, but only with its attribution, which therefore travels inside the artifact.

3 System and Deep Cases

RAY is three loosely coupled planes: a data plane (GitHub Actions pipelines through the harness, publishing hashed artifacts), a presentation plane (an installable React + MapLibre PWA, keyless and offline-cacheable), and an agent plane (any OpenAI-compatible endpoint wrapped in policy pre-check → grounded generation → claim post-check → audit). If the agent plane is down, the maps and analysis keep working — graceful degradation as fail-closed design made visible.

Tier 1 — nationwide watch. Four keyless federal connectors (USGS earthquakes, NWS alerts, NHC tropical cyclones, NIFC/WFIGS wildfires) refresh hourly in CI, writing append-only gzip snapshots and a national watch product. Per-source failures are declared, not hidden: the map badge reports *mapped-of-total* features and names sources that failed, and an ArcGIS error envelope fails the wildfire connector closed instead of reporting zero fires. NWS alerts render hollow — an advisory about what may come never looks like an occurrence. A Day-1 flash-flood outlook (NOAA/WPC, public domain) publishes as a *separate* product beside the watch, carrying its own declared boundary: outlook only, no damage conclusions, never a replacement for official warnings.

Tier 2/3 — three deep cases, one harness. Exposure, vulnerability, and damage analysis exist only inside three areas of interest, and the app says so rather than extrapolating:

- *Eaton Fire 2025 (CA):* 18,428 CAL FIRE DINS structure points → a 265-cell H3 r9 damage grid with per-severity counts and mandatory uncertainty; CDC SVI 2022 joined across 20 Census tracts with the downscaling approximation declared per feature; 2,244 reliability-gated cross-view street-imagery samples → a 109-cell evidence grid; 46,341 residents (2020 census blocks, centroid-allocated with the pre-event vintage declared) and 27 OSM critical facilities. A damage class with n=30 is declared as lacking statistical power rather than hidden.
- *Hurricane Milton 2024 (FL):* 2,556 labeled pre/post street-view pairs → a 15-cell grid at Horseshoe Beach, each feature carrying the attribution caveat that post-imagery is season-cumulative (Debby, Helene, Milton); 5,618 Pinellas County cells with county debris-program volumes and wind/rain covariates. 772,293 residents and 200 OSM facilities across the two AOIs. A generated-imagery dataset was excluded, and the exclusion itself is auditable in the event dossier.
- *Hurricane Ian 2022 (FL):* 886 matched samples → a 190-cell evidence grid; 4,121 further street-view positions ship as a *density-only* layer because no verifiable per-point severity link exists — candor over coverage; 5,428 residents and 79 OSM facilities.

All source imagery traces to a hashed dataset registry (134,272 files, ~33 GB, SHA-256). Facility context declares OSM *presence, never operational status*; population outside the evaluated tiles is declared as a total, never mapped into tiles the event has no evidence for;

and no claim-plane rule authorizes facility claims, so the agent's answers about them default-deny.

Two modes. Residents geocode an address and receive a plain-language dossier — stat callouts with their qualifiers attached, declared unknowns rendered as prominently as findings, and one of three coverage answers: covered, outside the evaluated areas, or *not determined* when a layer could not be read (the system will not claim an address is outside areas it failed to look at). Planners weight structural damage against social vulnerability on a live slider (every adjustment audit-logged), inspect lineage, and can draw an area and ask about it; the answer covers only the evaluated tiles the selection contains, never merges statistics across events, and highlights the tiles it cited.

4 Model Output as Governed Evidence

The obvious next step for any imagery-rich disaster system is to let a vision–language model (VLM) grade damage. The harness turns that step into a question: under what conditions may a model's guess become evidence? Our answer is mechanical. Model output is a distinct evidence class, `model_derived`, that carries its own measured accuracy alongside it or carries nothing; no claim-plane rule authorizes a claim from it; and the agent's evidence store refuses to load it. The numbers below are therefore published as an *evaluation of a method on labelled cases*, never as facts about places.

Setup. We ported the Damage Recognition prompts of RAPID [5] verbatim — Prompt C for a single post-event image on the five CAL FIRE DINS classes, Prompt B for a pre/post pair on three human-perception classes — and froze each as a `prompt_sha256` in every record, so a later edit is visible as a different prompt rather than a silent change in the numbers. The model is any OpenAI-compatible vision endpoint; here, open weights served by Ollama on one RTX 3090 (24 GB), so the run is keyless and repeatable from a workstation. Three cases: *Palisades 2025* (RAPID Dataset C2: 295 street-view images, folder-level DINS truth, all graded); *Milton 2024* (the Bi-Temporal pair set, 100 pairs per class, seed 2026); *Eaton 2025* (the matched cross-view set, 300 per class with the n=30 minority class fully included; Prompt C sees the post-event field image only — declared, not hidden — and its five classes are collapsed onto the set's three repairability classes by DINS semantics before scoring). Every prediction is a record: image sha256, truth, prediction, response digest, latency, model digest. An unparseable answer is recorded as unparseable, never coerced into a class. A stage aborts if more than 20 % of answers are off-schema, if a metric leaves $[0, 1]$, if fewer than 95 % of graded samples fall inside the grid the event already describes, or if any product lacks an uncertainty block. We report accuracy and RAPID's normalized cross-severity error, NCSE = $\overline{|y - \hat{y}|}/(K-1)$.

What the runs say. Table 1 reports seven models (a 7B reference plus six candidates), each committed with its own eval summary, prediction records, agreement grid, and GPU placement as it finished. Four things hold across the table. (i) *On a single post-event image, every open model beats the 7B reference, and the best is within two points of the paper's closed models.* Size is not the axis: gemma3:12b (0.546) is within one point of qwen3-vl:32b (0.554) and ahead of gemma3:27b. (ii) *On pre/post pairs, no open model beats the 7B reference, and the failures are systematic.* qwen3-vl:32b calls 258 of 300 pairs Severe (Mild recall 0.01); gemma3:12b calls 262 Moderate (Severe recall 0.17); mistral puts 213 in Moderate (Mild recall 0.13). Each model has a different fixed bias on the change task, so a vote across them would not be a fix; the per-class recall column in the committed table is the finding. (iii) *Eaton's small class is where every model is weak.* `damaged_repairable` (n=30) recall runs from 0.03 to 0.43 while the two large classes sit above 0.9; the headline 0.93 is the large classes. (iv) *Latency is a model property.* The qwen3 family spends its time in hidden reasoning before a 60-character answer: qwen3-vl:8b's median is 5 s, but nine Palisades images took over 30 s, and on Milton 24 pairs hit the client timeout and were recorded as unavailable (8.3 % unanswered, under the 20 % bound, so the case stands with its unanswered rate on the record).

Table 1: Six open VLMs, one pass at temperature 0, quantised weights, on the same seeded samples under the same prompt digests. Accuracy / NCSE. Blocks are not comparable across columns (different class counts). The closed-model row is RAPID's published number on the same Palisades images and on a 150-pair subset of the Milton set.

model	Palisades (5 cl.)	Milton pairs (3 cl.)	Eaton (3 cl.)
qwen2.5vl:7b (ref.)	0.475 / 0.213	**0.597** / 0.238	0.910 / 0.049
qwen3-vl:32b†	**0.554** / 0.176	0.381 / 0.431	0.901 / 0.053
qwen2.5vl:32b†‡	0.507 / 0.206	0.560 / 0.240	0.931 / 0.039
gemma3:27b	0.529 / 0.187	0.500 / 0.257	0.914 / 0.049
mistral-small3.2:24b	0.505 / 0.203	0.487 / 0.258	**0.935** / 0.037
gemma3:12b	0.546 / 0.178	0.427 / 0.287	0.906 / 0.051
qwen3-vl:8b	0.514 / 0.195	0.462 / 0.324	0.924 / 0.041
closed (RAPID)	0.573 GPT-5-mini	0.591 GPT-5.1	—

† 8k context so the weights fit the card; ‡ 95 % of the model in VRAM, the rest on CPU — recorded in the run, because a spilled model is a different run. A seventh model (llama3.2-vision:11b) could not be loaded by the serving stack and is declared as a serving failure, not scored.

What the harness did with it. The agreement grids sit beside the human-labelled grids they are evaluated against, flagged `model_derived`; the publication planner classifies them as evaluation products; the claim plane has no rule that admits them; and the gateway's evidence store skips them and records the exclusion. A reader can verify every number from the committed prediction records and prompt digests without a GPU. What no reader can do — and what the system therefore does not do — is turn a 0.93 into a sentence about a street.

5 The Harness Catching Its Own System

Three preserved incidents are, we believe, the strongest evidence the mechanism does real work.

The publication boundary. A parcel-level DINS file (18,428 points with per-structure damage) reached the public site through an asset-copying script, violating four written statements in the repository — while breaking zero policy rules, because the claim plane governs what the agent *asserts* and nothing had asserted the file. The fix was structural: a distribution plane in the same policy file, a publication-boundary planner, and a CI gate that verifies the assembled site; the public surface dropped from 30 files to 16. The incident report is committed alongside the code it indicts.

Citation by default, inverted. The claim post-check originally required citations only on sentences containing digits, so "*Your neighborhood was not significantly affected*" passed uncited. The check was inverted to citation-by-default with a closed exemption set; acceptance on a representative draft set fell from 9/13 to 7/13 sentences — the two newly refused sentences being exactly the uncited reassurances a resident should not receive.

The agent reading what the publisher withheld. The gateway's evidence store loaded every event dossier and grid on disk, ignoring both the `published_events` list and the `model_derived` flag. The first evaluation case written under `events/` — the Palisades VLM run of Section 4 — would have been located by point and its predictions cited as tile facts, and Milton's model-graded pair grid would have been served beside the human-labelled grid it is evaluated against. The publication planner had excluded them; the agent had not. The fix makes both planes publish from one list, has the store skip model-derived products and record every exclusion, and refuses to start a store over "every event on disk" as a misconfiguration rather than a fallback (20 new tests). The pattern is the same as the first incident: a constraint written in one place and enforced in another is enforced nowhere.

6 Evaluation Environment

Following Yang and Zou [6], the test suite is designed as the paper's verifiable evaluation environment rather than an afterthought: 363 Python tests and 67 app tests run in CI on every push, including a cell-by-cell policy-matrix sweep (every role × tier × resolution combination against expected authorization), adversarial claim-check cases, fail-closed connector tests against real error envelopes, the containment property test for content-free lookup records, the gateway-scope tests above, and the VLM stages' parsing, fail-closed and resume logic with the model call injected. Anchor checks resolve every path-shaped reference in the documentation against the repository, so the manual cannot silently rot. A CI check regenerates the publication allowlist from the policy and fails on any drift, which is how a hand edit cannot widen the public surface. Judges can run the environment in two commands from the README.

7 Related Work

Autonomous-GIS agents demonstrate that LLMs can operate spatial tooling [4, 7]; our contribution is orthogonal — a harness that constrains any such agent's claims to its evidence. RAPID [5] shows that a multi-agent pipeline over closed APIs can grade damage from imagery interpretably; we keep its prompts, metric and acceptance rules as harness stages, replace the closed models with open ones, and decline its LLM task planner, because a model choosing which agents run conflicts with a policy plane. Operational platforms such as the Texas Disaster Information System [3] field forecast-forward impact summaries at state scale; accountability there rests on interface disclaimers, whereas RAY binds each sentence to a hashed artifact and a policy rule. Global registries (EM-DAT [1], GDACS [2]) record that events happened; none carries, per row, what resolution of claim the evidence authorizes or whether a reader may keep it — the two fields this system computes for every artifact.

8 Limitations

The deep-case builders read a ~33 GB corpus on the maintainer's workstation, so third parties can verify the committed artifacts, hashes, and audit logs but not regenerate them. The VLM results are one pass at temperature zero on 4-bit quantised weights; a model digest pins the weights, not the arithmetic, and 32B is the ceiling of a 24 GB card. Milton and Eaton are seeded stratified samples, and the closed-model reference for Milton was graded on a different 150-pair subset. The agent gateway runs locally (no hosted endpoint; no auth or rate limiting yet), so the public demo ships without chat. The re-derivable regime is implemented and tested against a stub but has never run against a live commercial API. Nationwide coverage is Tier-1 watch only; the app declares this boundary instead of smoothing over it, which we consider a feature, but reviewers should not mistake three deep cases for national analysis capability.

Acknowledgments

RAY is an MIT-licensed research prototype. It is not an official forecasting or warning service.

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